Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/25/2020
PI Name:	DeChurch, Leslie Ph.D.		
Project Title:	SCALE: Shared Cognitive Architectures for Long-term Exploration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Previously at Georgia Institute of Technology until July 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
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No. of PhD Candidates:	6	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Contractor, Noshir Ph.D. (Northwestern University) Johnson, Jeffrey Ph.D. (University of Florida, Gainesville)		
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Task Description:	NOTE: Continuation of "SCALE: Shared Cognitive Architectures for Long-term Exploration," grant NNX15AM26G, due to Principal Investigator move to Northwestern University from Georgia Tech in Fall 2016. Among the remarkable team challenges NASA faces in long distance space exploration (LDSE) missions is the need to maintain team shared mental models (SMMs). Maintaining team SMMs requires the ability to detect shifts in cognition that will likely occur during the mission that could lead to ineffective crew functioning and performance. Maintaining team SMMs also requires validated countermeasures for bringing team members' cognitive understanding back into alignment. Leaving low Earth orbit is extreme teamwork -- team SMMs need to be maintained within teams operating close up (the crew), and between teams operating at an unprecedented distance (i.e., the crew & ground; 33 million miles in the case of a Mars Mission). A multidisciplinary research team will leverage expertise in Psychology, Industrial Engineering, & Anthropology to understand the emergence and outcomes of critical shifts in team cognition over LDSE missions. What are the triggering events of SMM divergence, how can we detect them, and which countermeasures most effectively bring them back into alignment? This project leverages a novel conceptual framework of shared cognitive architecture (SCA) to understand the patterns of SMMs that dynamically link members of teams, and teams to other teams, as they go beyond low Earth orbit. We use semantic analysis to identify cognitive shifts, and relational event network analysis to understand the antecedents and consequences of these shifts. We use these alongside an agent-based model fit on LDSE analogue data, so that we can explore an exhaustive set of potential triggering conditions that must be unpacked to conduct efficient ground analogue research. We then conduct this research in HERA (Human Exploration Research Analog), Moonwalk, and Antarctica. The project culminates in the evaluation of a dashboard fed with the results of computational modeling, human validation, and lexical markers to detect and suggest countermeasures to maintain SMMs through time and space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many of the environmental conditions faced by crews during LDSE missions are not unique to space travel. With our research program, we hope to reveal how conditions seen in organizations across the world impact shared cognitive architecture, and what we can do to mitigate these risks. Our meta-analysis has begun to investigate the impact of shared cognitive architecture on team outcomes across a variety of environmental conditions seen in LDSE missions that can also be applied to work in all organizations.
Task Progress:	[Ed. note: Report submitted in September 2018 to Human Research Program personnel; Task Book received fall 2019] Among the significant team challenges NASA will face during long distance space exploration is the need to maintain shared cognition within the space crew and among the system of teams that support the crew from Earth (i.e., the space exploration multiteam system). Shared cognition is defined as team members' shared understanding of knowledge regarding the team's environment, tasks, expertise, roles, and responsibilities. Shared cognition needs to be maintained within teams who operate at a short distance (i.e., the crew), and also between teams who operate at a great distance (i.e., the crew and ground). Research suggests the extent to which cognition is shared within the team and the broader multiteam system has implications for team process, performance, affect, and viability over time. Over the course of this three-year programmatic research program, including previous work under grant NNX15AM26G at Georgia Tech, we have three major foci of activity: (1) to develop and test ways to measure shared cognition in teams and multiteam systems, (2) to explore the antecedents, consequences, and moderating conditions associated with shared cognition, and (3) to model shared cognition in teams and within multiteam systems so as to inform interventions. In Year 2, we made significant strides in each of these areas with six major projects that speak to one or more of the above foci. In Study 1, we use meta-analysis to explore the conditions under which shared cognition is most important to crew performance. We explore three types of moderating variables, including team type (e.g., team diversity, autonomy, and lifespan), team context (e.g., virtuality, temporal and geographic dispersion, and organizational embeddedness), and team conditions (e.g., physical and psychological stressors). Results help us determine (1) the extent to which prior teams research is relevant to long distance space exploration, (2) factors which need to be modeled effectively in order to predict, diagnose, and prevent breakdowns in shared cognition, and (3) which antecedent and moderating conditions are most relevant to cognitive sharedness in long distance space exploration. In Study 2, we developed an approach to using text analytics as an unobtrusive means of measuring shared cognition using the archived mission transcripts from the Skylab missions. We identified the text-based signatures of cognitive breakdown that ultimately led to conflict in Skylab 4. We also explored the dynamic conversational patterns in each mission that were indicative of cognitive sharedness or its breakdown. This study informs our broader goal of developing efficient and unobtrusive measures of shared cognition in multiteam systems. In Year 3, we expanded this analysis to examine the reasons why the cognitive breakdown was not averted earlier. We refined our previous analysis to look at the specific subject areas where understandings diverged and the characteristics of these subjects. Our findings suggest that discussion of scheduling, morale, and stressors were highly constrained by the medium and norms of communication and this constraint hindered the development of shared understandings around these issues. In Study 3, we programmed the agent based model (ABM) of crew member mental models, and estimated parameters using empirical data collected from the HERA analog. We developed the ABM in two phases. First, we canvassed the relevant literature on shared cognition to identify its antecedent conditions. Second, we collected laboratory and analog data to estimate and then test the parameters of the agent based model. In particular, we included personality factors, social relations, situational factors, and task stream characteristics that associate with cognitive sharedness. In year 3, we implemented the model in the NetLogo ABM platform. Our model is essentially utilizing data on individual, relational, situational, and contextual factors at each point in time in the mission to predict crew mental models similarity. The model takes each crew member's initial mental model, then considers who interacts with whom, which tasks various members work on, etc., to re-compute the mental model, and then to compute the similarity of each crew member's model with those of their crewmates, at each tick in the schedule. The agent-based model estimates the probability that individual i mental model changes from time t to time t+1. Then, we used BehaviorSearch tool to estimate the parameters that best fit the computational model to the empirical data. Studies 4-5 speak directly to our major goal of identifying antecedents conditions of cognitive sharedness. Both utilize data gathered in HERA C3 and C4. In Study 4, we collected data in the HERA analog using daily planning surveys that elicit crew members goals and challenges. In study 5, we used Project RED relay data to understand the nature of communication errors in large, distributed, space exploration systems. Our research will provide fundamental, generalizable findings in the area of shared cognition in teams and multiteam

	systems. Researchers have bemoaned the inefficiencies associated with measuring shared cognition as historically such measurements have been intrusive and time-consuming. In this program of research, we are exploring the use of text analytics, social role network analyses, and surveys as efficient and less intrusive means of assessing cognitive sharedness within the team and broader multiteam system. Second, as teamwork becomes more prevalent across workplaces and occupations and its nature more variable, a broader array of antecedent, moderating, and mediating conditions associated with shared cognition come into play. We are conducting meta-analytic, archival, experimental, and analog studies designed to identify and understand the antecedents and consequences of shared cognition in teams and multiteam systems. Finally, breakdowns in shared cognition must be identified and addressed quickly before team functioning is irrevocably affected. We use data collected in these meta-analytic, archival, experimental, and analog studies to develop an agent based model which will be used in Year 4 to conduct virtual experiments and ultimately create a dashboard for diagnosing and developing interventions for breakdowns in shared cognition. This research will be useful for investigating shared cognition in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN--European Organization for Nuclear Research), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems. In this review period, we have collected data across multiple platforms to begin to understand the antecedent, consequent, and moderating conditions associated with shared cognition. In addition to providing Earth Benefits in the area of shared cognition in teams and multiteam systems, we are also testing our hypotheses in analogs that recreate the unique contexts in which astronauts operate (i.e., extreme, isolated/confined environments). We will compare results across research paradigms to see how effects of isolation/confinement may moderate shared cognition's role in team functioning, thus providing unique findings that will inform NASA's operations aboard the International Space Station (ISS) and on Long-Duration Space Exploration (LDSE) missions.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
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